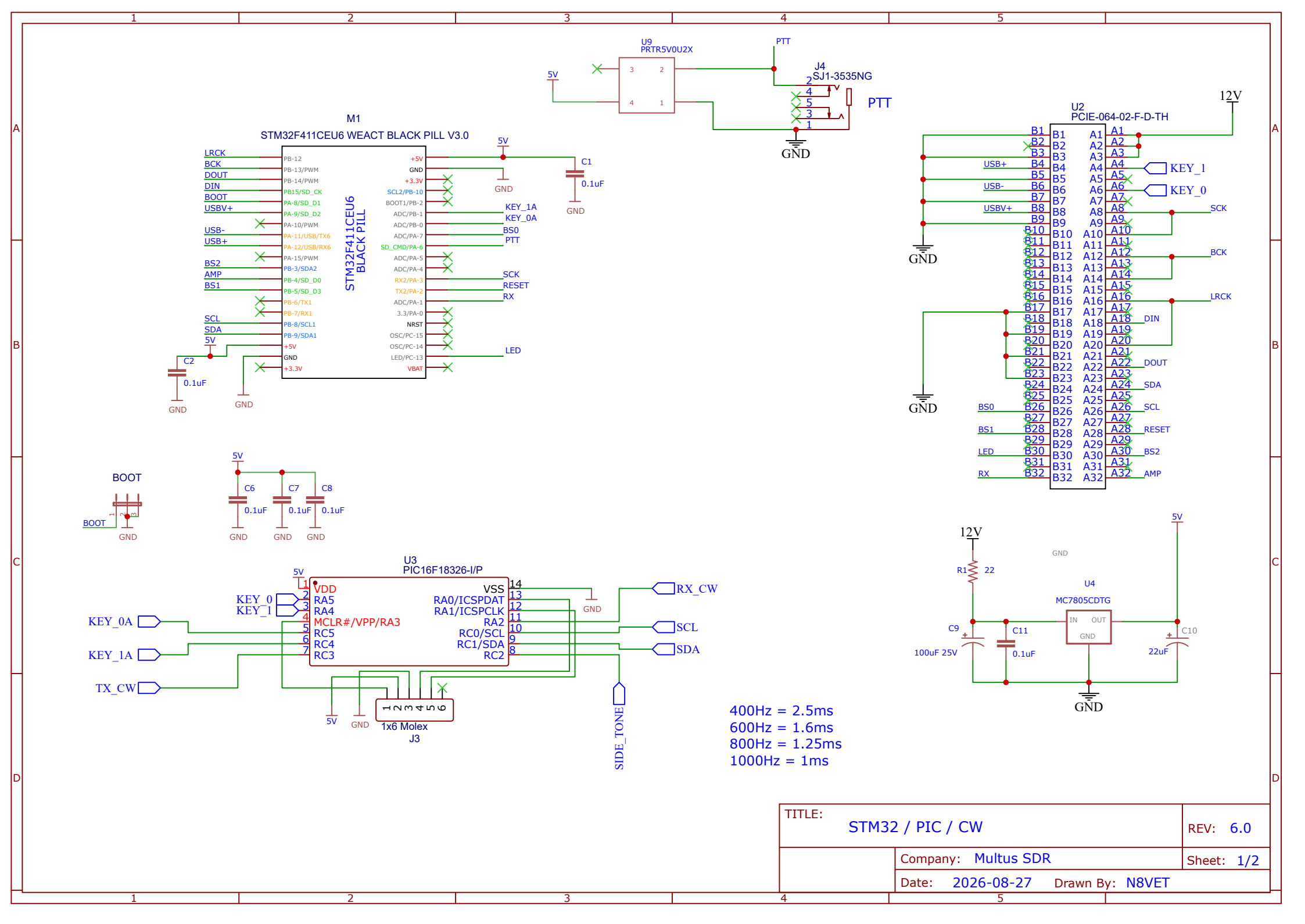
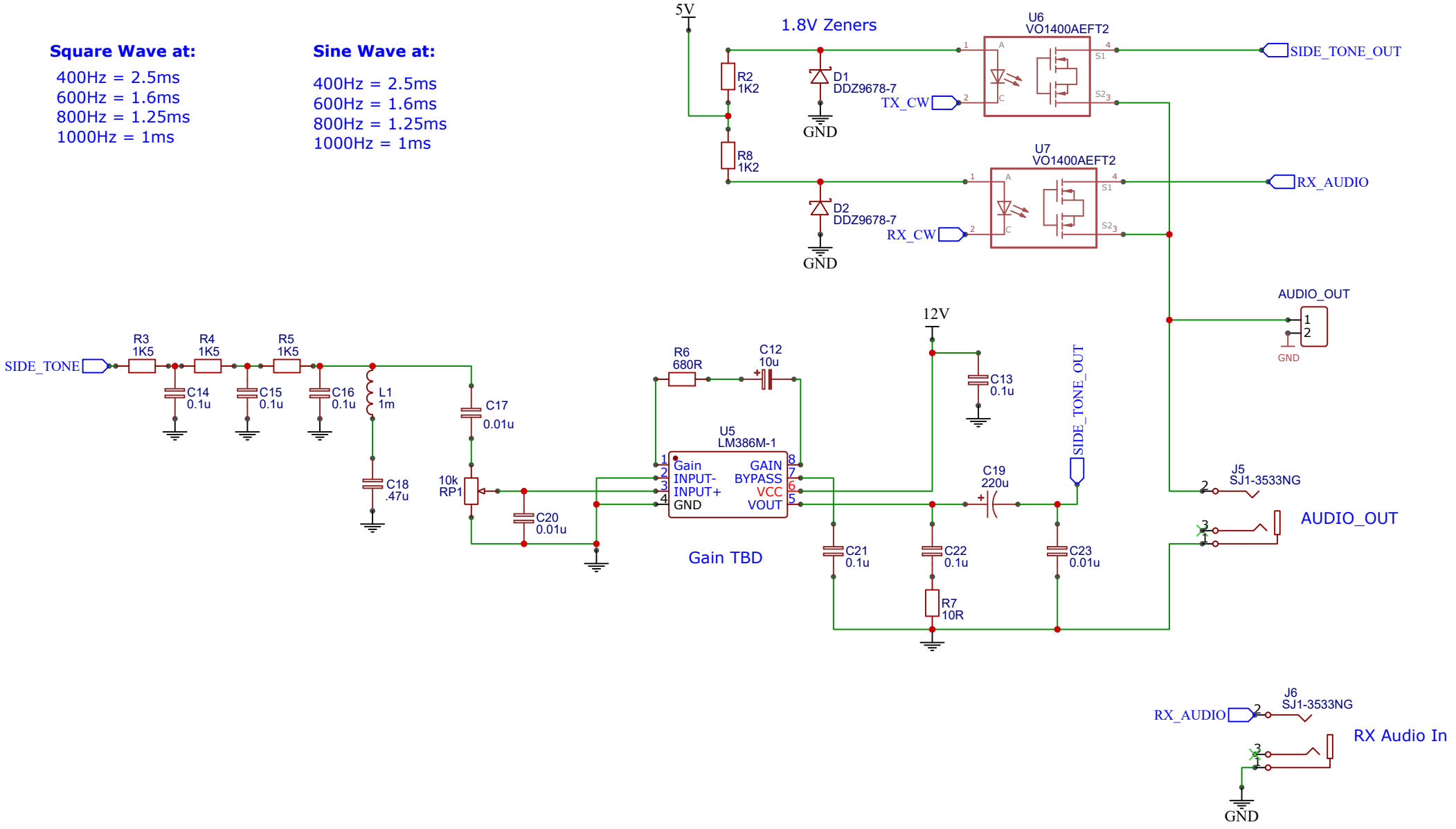

U9
PRTR5V0U2X
PTT
5V
J4
SJ1-3535NG
PTT
GND
M1
STM32F411CEU6 WEACT BLACK PILL V3.0
LRCK
BCK
DOUT
DIN
BOOT
USBV+
USB-
USB+
BS2
AMP
BS1
SCL
SDA
5V
PB-12
PB-13/PWM
PB-14/PWM
PB15/SD_CK
PA-8/SD_D1
PA-9/SD_D2
PA-10/PWM
PA-11/USB/TX6
PA-12/USB/RX6
PA-15/PWM
PB-3/SDA2
PB-4/SD_D0
PB-5/SD_D3
PB-6/TX1
PB-7/RX1
PB-8/SCL1
PB-9/SDA1
+5V
GND
+3.3V
STM32F411CEU6
BLACK PILL
+5V
GND
+3.3V
SCL2/PB-10
BOOT1/PB-2
ADC/PB-1
ADC/PB-0
ADC/PA-7
SD_CMD/PA-6
ADC/PA-5
ADC/PA-4
RX2/PA-3
TX2/PA-2
ADC/PA-1
3.3/PA-0
NRST
OSC/PC-15
OSC/PC-14
LED/PC-13
VBAT
KEY_1A
KEY_0A
BS0
PTT
SCK
RESET
RX
LED
C1
0.1uF
C2
0.1uF
U2
PCIE-064-02-F-D-TH
12V
USB+
USB-
USBV+
KEY_1
KEY_0
SCK
BCK
LRCK
DIN
DOUT
SDA
SCL
RESET
BS2
AMP
BS0
BS1
LED
RX
BOOT
C6
C7
C8
0.1uF
U3
PIC16F18326-I/P
KEY_0
KEY_1
KEY_0A
KEY_1A
TX_CW
VDD
RA5
RA4
MCLR#/VPP/RA3
RC5
RC4
RC3
VSS
RA0/ICSPDAT
RA1/ICSPCLK
RA2
RC0/SCL
RC1/SDA
RC2
RX_CW
SCL
SDA
SIDE_TONE
1x6 Molex
J3
R1
22
U4
MC7805CDTG
IN
OUT
C9
100uF 25V
C11
0.1uF
C10
22uF
400Hz = 2.5ms
600Hz = 1.6ms
800Hz = 1.25ms
1000Hz = 1ms
TITLE:
STM32 / PIC / CW
REV: 6.0
Company: Multus SDR
Sheet: 1/2
Date: 2026-08-27
Drawn By: N8VET

Square Wave at:

400Hz = 2.5ms
600Hz = 1.6ms
800Hz = 1.25ms
1000Hz = 1ms

Sine Wave at:

400Hz = 2.5ms
600Hz = 1.6ms
800Hz = 1.25ms
1000Hz = 1ms



TITLE: Side Tone Generator		REV: 6.0
	Company: Multus SDR	Sheet: 2/2
	Date: 2026-08-27 Drawn By: N8VET	